

National University of Computer and Emerging Sciences
Lahore Campus

Theory of Automata (CS3005)

Final- Exam

Date: Jan 3rd, 2026

Course Instructor(s)

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Total Time (Hrs): 3

Total Marks: 90

Total Questions: 5

Roll No

Section

Student Signature

Do not write below this line

- Attempt all the questions. No marks without working.
- In case of confusion or ambiguity, make a reasonable assumption.
- Solve Q2 till Q5 on the question paper. Show working on the answer sheet (MUST). Solve Q1 on the answer sheet.
- Provide only one final solution to each question. Submitting multiple solutions will result in cancellation, and no marks will be awarded.

Final marks on flex set →

CLO	1	2	3	4		Total
Question#	Q2	Q4	Q3	Q1	Q5	
Marks (total)	15	15	15	20	25	90
Marks (obtained)						

CLO 4: Demonstrate, differentiate, and manipulate formal descriptions of languages, automata, and grammars with a focus on non-context-free languages using Turing Machines.

Q1: [15+5] Design a 3-tape Turing Machine that converts a unary number on Tape 1 (e.g., 111111) into its binary representation on Tape 3 (e.g., 110).

Tape 2 may be used as a working area. You are allowed to modify Tape 1 during computation, but the final binary result must appear on Tape 3. **[15 marks]**

(Note: Write the algorithm in 3–4 precise steps, avoiding unnecessary explanation. **[5 marks]** Calling predefined Turing Machines from class or textbooks are not allowed; all components must be designed from scratch.

Example conversions: • 11 → 10 • 11111 → 101 • 11111111 → 1000 • 11111111111 → 1011	Tape configuration: • Tape 1: Input unary number • Tape 2: Working tape • Tape 3: Final binary output
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The output may contain leading zeros; for example, an input of 11111 may produce outputs such as 101 or 00101, both of which are valid.

218 binary addition.

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CLO 1: Identify formal language classes and prove language membership properties. Prove and disprove theorems establishing key properties of formal languages and automata

Q2: [15 marks] Decide, using the pumping lemma, whether the following language is a CFL or not. Provide a PDA (if it is).

$\{x \in \{a, b\}^* \mid |x| \text{ is even and the first half of } x \text{ has one more } a\text{'s than the second half}\}$

$u = vwxy^mz$

n for $\forall$
$u \in L$
$ u \geq n$

Four cases:

① $|wxy| > 0$

② $|wxy| \leq n$

③ $\forall, m \geq 0, vwxy^mz \in L$

Start with assumption that language is CFL.

Then after contradiction, and proof with different strings and scenarios we will say that

① lang. is Non-CFL.

② No PDA.

① if $|u|$ is odd $\notin L$

Else

(a) First half has all a's
— second half b's or less a's

(b) First half all b's —
Second half a's or less a's

(c) In second half - 'a' cond.

(d) In second half - 'b' cond.

→ NOT a CFL so
no PDA.

CLO 3: Differentiate and manipulate formal descriptions of languages, automata, and grammars with a focus on context-free languages using automata (PDA and NPDA).

6 - normal
2 - all a's
2 - all b's

Q3: [15 = 10 + 5 marks] Develop CFG for the following language with $\Sigma = \{a, b, c, d\}$.
 $L = \{c^i a^k b^m d^j \mid \text{where } [j = 2i \text{ and } k \neq m]\}$.

Develop CFG in phases:

Phase 1: CFG for $a^k b^m$

$S \rightarrow aSb / A / B$
 $A \rightarrow aA / a$
 $B \rightarrow Bb / b$

$\begin{pmatrix} abbb \\ aaab \end{pmatrix}$
Valid

$k \neq m \Rightarrow k < m \text{ or } k > m$

$[a, b]$

valid

abb	aab
aabbb	aaabb
aaa	bbb

invalid

ab	aabb	ba	bbba
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$(S \rightarrow \lambda) \times$

Phase 2: CFG for L

$L \rightarrow cLdd / s / cSdd$

$c^i d^{2i}$
 odd

$\times [L \rightarrow \lambda]$

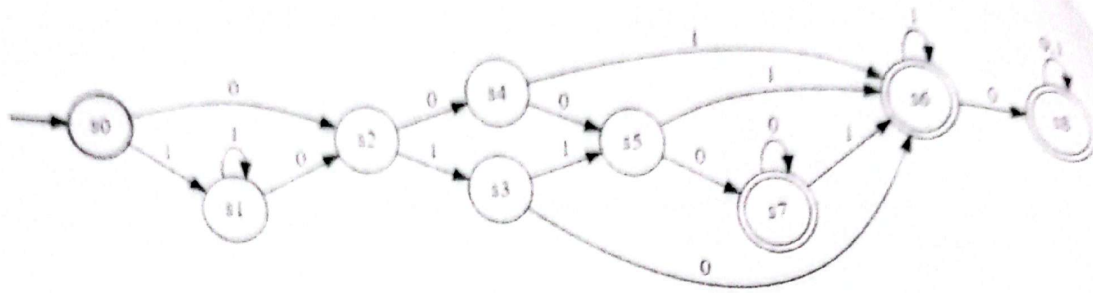
$cabbdd$
 $ccjddddd$
valid cases of a/b

CLO 2: Differentiate and manipulate formal descriptions of languages, automata, and grammars with a focus on non-regular and regular using automata (DFA, NFA, NFA-NUL).

Q4: [15 marks] For the given nine-state DFA, apply the state minimization algorithm to reduce the number of states without affecting the language being accepted by the DFA.

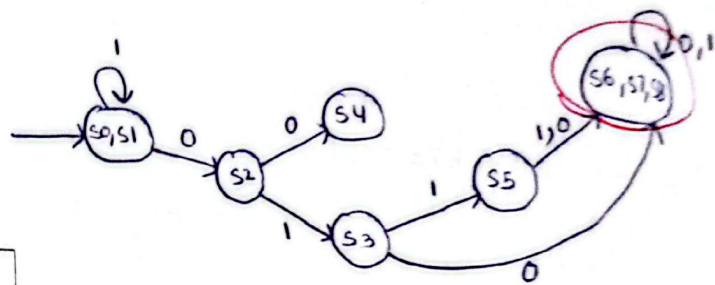
- Fill the table formally. [10 marks]
- Draw a minimized DFA. [5 marks]

VOTE: All sections must fill the table except 5A, 5B, and 5K. Mentioned sections will solve part (a) on the answer booklet, as discussed in class, and part (b) in the space given below.



Draw Minimized DFA

	S0	S1	S2	S3	S4	S5	S6	S7
S1	-							
S2	3	3						
S3	2	2	2					
S4	2	2	2	2				
S5	2	2	2	2	2			
S6	1	1	1	1	1	1		
S7	1	1	1	1	1	1	-	
S8	1	1	1	1	1	1	-	-



✓ 15

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Q4: Demonstrate differentiate and manipulate formal descriptions of languages, automata and grammars with focus on non context free languages using Turing Machines

Q5: [20+5 marks] Given the Turing Machine, perform a dry run for the specified input. Each time the machine enters **state A**, record the **contents of the tape** and the **current position of the tape head** (by **underlining it**) in the space provided below. **Clearly label each row to indicate whether it corresponds to state A [20 marks]**. Use only the cells provided below that are required. If additional space is needed, use the last page

Input String: \$aba@ababa initially the head is at \$

Sr.	Content of the tape and head	
1	\$ a b a @ a b a b a Δ Δ Δ Δ	2
2	\$ <u>a</u> b a @ X b a b a Δ Δ Δ	4.5
3	\$ <u>a</u> b a @ X y a b a Δ Δ Δ	4.5
4	\$ <u>a</u> b a @ X y X b a Δ Δ	4.5
5	\$ <u>a</u> b a @ X y X y a Δ Δ	4.5
6	\$ A B a @ X y X y X Δ Δ	

0.5 for header

Q5. (20+5 marks) Given the Turing Machine, perform a dry run for the specified input. Each time the machine enters state A, record the contents of the tape and the current position of the tape head (by underlining it) in the space provided below. Clearly label each row to indicate whether it corresponds to state A [20 marks]. Use only the cells provided below that are required. If additional space is needed, use the last page.

Input String: Saba@ababaΔΔ initially the head is at 5

all working is shown on answer sheet

Sr.	Content of the tape and head
1.	\$ ↓ a b a @ a b a b a Δ Δ
2.	\$ ↓ a b a @ x b a b a Δ 1
3.	\$ ↓ a b a @ x y a b a Δ 1
4.	\$ ↓ a b a @ x y x b a Δ 1 1
5.	\$ ↓ a b a @ x y x y a Δ 1 1
6.	

Final Content of Tape and pointer [2 marks]

\$	A	B	a	@	x	y	x	y	x	↓ △	1	1							
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has

Explain in 3 to 4 lines what the TM is doing? [3 marks]

The TM is accepting the strings that are same when read from front or back and odd length.

(2)

(24)